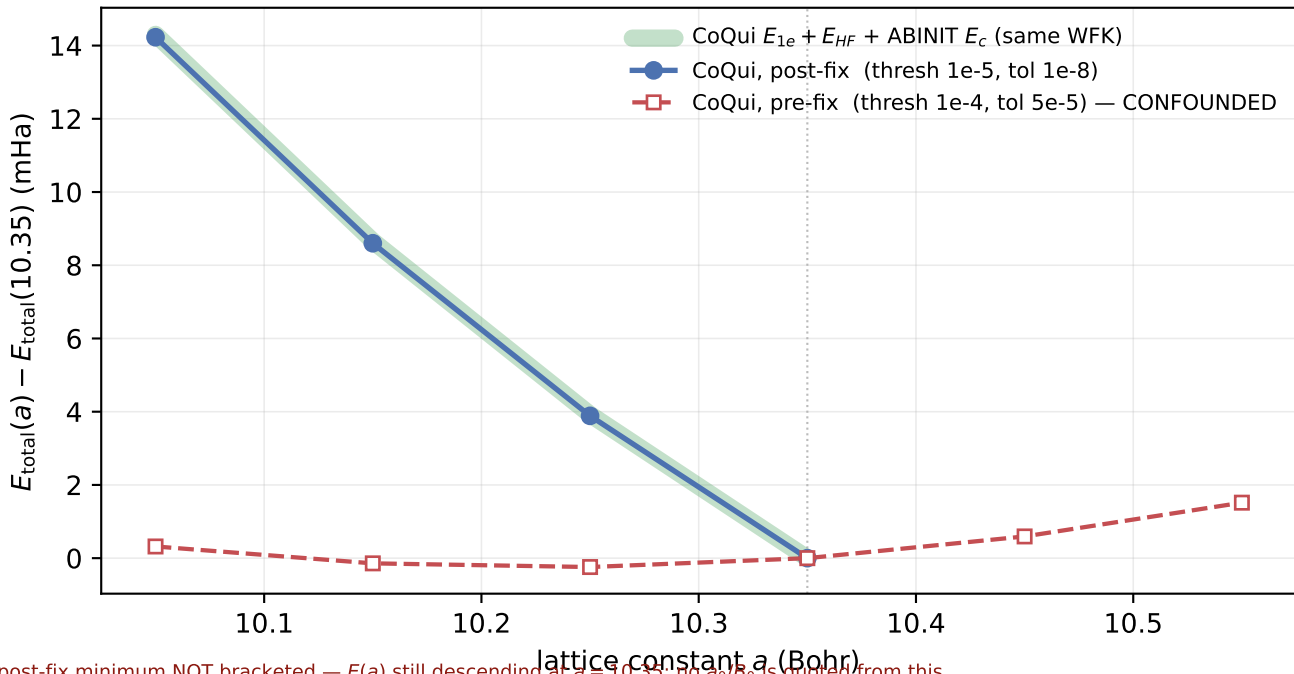


# Si RPA@PBE equation of state, $n_{\text{band}} = 500$



post-fix minimum NOT bracketed —  $E(a)$  still descending at  $a = 10.35$ ; no  $a_0/B_0$  is quoted from this.  
 pre-fix curve is at DIFFERENT settings: its  $E_{1e} + E_{HF}$  alone differs from post-fix by 6.15 mHa across 0.30 Bohr (monotone), so pre-vs-post here is NOT a clean measure of the  $V_{LL}$  fix — use Fig. 2 ( $E_c$ ).  
 The pre-fix curve is nearly FLAT because its over-steep  $E_c$  ( $4\times$  ABINIT's volume dependence) happened to cancel the  $E_{1e} + E_{HF}$  slope. Its  $a_0 \approx 10.23$  was error cancellation, not agreement.

Total = CoQui( $E_{1e} + E_{HF} + E_{RPA}$ ) +  $E_{Ewald}$ (ABINIT, Ha). ABINIT reports no RPA total in this convention; the green band substitutes its  $E_c$  into CoQui's decomposition on the identical WFK. All curves anchored at  $a = 10.35$  Bohr (dotted line).